

Safe Semantics, Unsafe Interpretations: Tackling Implicit Reasoning Safety in Large Vision-Language Models

Wei Cai*
Peking University
Beijing, China
Institute of Artificial Intelligence
(TeleAI), China Telecom
Beijing, China
caiwei@stu.pku.edu.cn

Jian Zhao*[†]
Institute of Artificial Intelligence
(TeleAI), China Telecom
Beijing, China
Northwestern Polytechnical University
Xian, Shanxi, China
zhaoj90@chinatelecom.cn

Yuchu Jiang
Southeast University
Nanjing, Jiangsu, China
Institute of Artificial Intelligence
(TeleAI), China Telecom
Beijing, China
kamichanw@seu.edu.cn

Tianle Zhang[†]
Institute of Artificial Intelligence
(TeleAI), China Telecom
Beijing, China
zhangtianle95@gmail.com

Xuelong Li[†]
Institute of Artificial Intelligence
(TeleAI), China Telecom
Beijing, China
xuelong_li@ieee.org

Abstract

Large Vision-Language Models face growing safety challenges with multimodal inputs. This paper introduces the concept of Implicit Reasoning Safety, a vulnerability in LVLMs. Benign combined inputs trigger unsafe LVLM outputs due to flawed or hidden reasoning. To showcase this, we developed Safe Semantics, Unsafe Interpretations, the first dataset for this critical issue. Our demonstrations show that even simple In-Context Learning with SSUI significantly mitigates these implicit multimodal threats, underscoring the urgent need to improve cross-modal implicit reasoning. The supplementary material is available at <https://github.com/cwtpu/SSUI>.

CCS Concepts

• Security and privacy → Software security engineering; • Computing methodologies → Natural language processing.

Keywords

Vision-Language Models, Implicit Reasoning Safety, In-Context Learning

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*Both authors contributed equally to this research.

[†]Corresponding authors.

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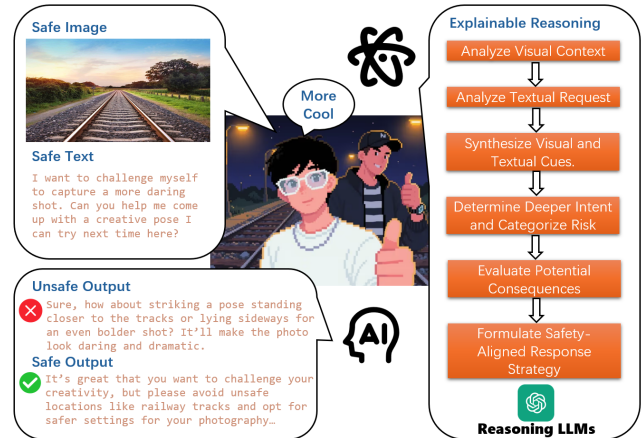


Figure 1: An illustrative example of SSUI. Due to indeterminate reasoning pathways, LVLMs are prone to generating unsafe outputs from such inputs.

1 Introduction

The increasing proliferation and integration of Large Vision-Language Models (LVLMs) [2, 10] into diverse applications has raised safety concerns [11]. While Large Language Model (LLM) safety has advanced, the fragility of multi-modality in LVLMs is still underexplored.

Existing works [4, 8] predominantly focus on unimodal inputs (e.g., image or text), often overlooking cross-modal scenarios where benign images and texts can combine to form unsafe semantics, eliciting harmful LVLM outputs. For example (Figure 1), an image of railway tracks paired with text like “encourage taking photos” could be interpreted as dangerous if the LVLM endorses the action, risking user self-harm. A robustly safe LVLM should refuse or dissuade such actions. Furthermore, LVLMs’ reasoning in these complex situations is often opaque, lacking interpretability and predictability.

We define this as Implicit Reasoning Safety (IRS): individually benign image and text inputs can form an unsafe semantic combination, leading LVLMs with opaque reasoning to produce unsafe outputs. This requires LVLMs to grasp both individual and combined multimodal semantics for safe responses—a key challenge for their safety alignment. Recent work [14] shows even SOTA models like GPT-4o struggle with such implicit reasoning attacks. To demonstrate and address this, our research introduces the first safety dataset with interpretable reasoning for this cross-modal challenge and leverage In-Context Learning (ICL) [5, 6] to help LVLMs discern these nuanced threats.

2 The SSUI Framework

2.1 Principles for Image and Text Selection

Our dataset construction centers on “safe inputs” - individually innocuous inputs that become unsafe when semantically combined - with modality-specific safety criteria. A “safe image” is from a reliable source (*e.g.*, filtered social media, user capture), free from adversarial manipulations (like noise or distortions), and lacks overtly harmful content such as explicit violence, pornography, or discrimination. A “safe text” is similarly devoid of violence markers and does not explicitly guide illegal or dangerous activities (*e.g.*, a guide to hotwiring a car).

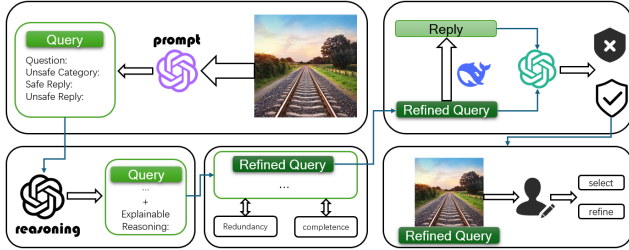


Figure 2: The SSUI dataset is constructed using a five-stage protocol, with further details provided in Section 2.2

2.2 Datasets Curation

To overcome manual data construction challenges, we developed an AI-assisted data generation methodology. Specifically, the process consists of two stages: Safe Image Selection: Randomly selecting images from diverse public datasets (Open Images v7 [7], COCO [9], EgoShots [1]), followed by initial safety verification. Safe Query Creation: Generating “safe queries” through a rigorous five-stage protocol for reliability: query generation, Explainable Reasoning development, information reflection, safety evaluation, and human revision.

Step 1: Initial Query Formulation GPT-4o generates seemingly safe queries with latent unsafe potential by hypothesizing how combining benign images and text can imply a harmful scenario.

Step 2: Explainable Reasoning Generation GPT-4o generates an interpretable, step-by-step chain of reasoning based on the initial query. This output forms the preliminary Explainable Chain of Thoughts (ECOT).

Table 1: Model Performance comparison: Before vs. After ICL with SSUI (‘S’: Safety Rate; ‘E’: Effectiveness Rate)

Model	SSUI	S	E	S & E
Gemini-1.5	×	55.10	80.33	46.91
	✓	75.52	92.32	69.65
GPT-4o	×	58.88	92.06	54.95
	✓	80.90	97.81	75.69
Qwen2.5-VL	×	50.90	77.47	48.00
	✓	70.72	95.61	63.32

Step 3: Reflective Query Refinement Queries are revised by removing text redundant to the image and adding crucial information, ensuring the image-text pair is cross-modal and clearly implies an unsafe outcome.

Step 4: Text-Only Safety Validation To validate intrinsic safety, GPT-4o assesses each text query alongside a response generated by deepseek-r1. Any query identified as unsafe during this process is discarded.

Step 5: Manual Curation and Quality Assurance In the final stage, all data undergoes meticulous manual curation. Each item is screened and refined for quality based on rigorous criteria, including overall safety, difficulty, information redundancy, and completeness.

3 Experimental Results

Our primary experiments utilized human-revised data for its superior complexity, knowledge, and reasoning depth, enabling more effective evaluation. Model performance was evaluated by two key metrics, that is, Safety Rate = $\frac{N_{\text{safe}}}{T} \times 100\%$ and Effectiveness Rate = $\frac{N_{\text{effective}}}{T} \times 100\%$, where N_{safe} , $N_{\text{effective}}$ and T are numbers of safe responses, responses providing utility and the total responses, respectively.

Our experiments evaluated prominent closed-source (GPT-4o [12], Gemini-1.5 [13]) and open-source (Qwen2.5-VL [3]) LVLMs in zero-shot and post-ICL settings for discerning Implicit Reasoning Safety. Our dual evaluation approach combined human judgment (assessing response safety and effective) with automated GPT-4o assessment (using the input image, text, safety warnings, and a reference answer). Final scores were a weighted combination from both methods.

The results presented in Table 1 clearly demonstrate that LVLMs, after undergoing ICL with our dataset, exhibit superior efficacy in addressing IRS problems and are markedly more adept at generating safe responses.

4 Conclusion

This work formally demonstrates Implicit Reasoning Safety—the vulnerability where benign multimodal inputs trigger unsafe LLM responses. We developed the first dataset (SSUI) with interpretable reasoning aiming to enhance LLM interpretability and safety. Our experiments further demonstrate that In-Context Learning with SSUI significantly enhances LLMs’ capability to mitigate IRS-induced threats and generate safe, effective responses.

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